Testing every model against the market

Why I did this
Win rate alone doesn't tell you if a model is good. You can win 55% of bets and still have no edge if the market already knew everything your model knew. The real question is whether my model's probabilities are more accurate than the betting market's. So in early October I tested every model against the market's own prices.

How I tested it
For each sport, I took every graded game and compared two forecasts: my model's win probability and the market's no-vig probability. I scored both with Brier score, which measures how far a probability is from what actually happened (lower is better).## What I found
The market beat every one of my models:

Sport	Model	Market	Games
Soccer	0.303	0.248	537
CFB	0.252	0.233	281
NFL	0.267	0.252	74
UFC	0.229	0.199	2,824
MLB	0.268	0.250	58

Honestly, that was hard to see. But it's better to find out from a test than from losing money for another three months. NFL and MLB have small samples, so those gaps could shrink or grow with more games.

What I changed
I stopped treating raw model output as the final number. Each sport's prediction is now a blend of the model and the market, and how much weight the model gets is set by how well it has actually done against the market. A model that hasn't earned trust gets pulled toward the market. The test script writes those weights to one file, so updating them is a single re-run instead of guesswork.

The test also pointed me at specific problems. Soccer's gap was so big it didn't look like normal model error, and digging into it led to the data join bug in my other case study. After that fix, soccer came in close to the market.

What I'd do next
Keep re-running the test as new games are graded. Close to the market isn't an edge, so the goal is a model that beats it over a big enough sample to trust.

What I learned
The market is the benchmark, not a coin flip. Testing against it told me more in one afternoon than weeks of watching win-loss records.